GLOWTAP

Database Performance

Local PostgreSQL · 4/8/2026, 3:11:58 am IST · 5 measured rounds · DB 5472 MB

26 workloads | 25 excellent | Latency suite | Health score 99%

TESTS	EXCELLENT	GOOD	ACCEPTABLE
26	25	1	0
timed queries			

SLOW	DB SIZE	LIVE ROWS	CPU CORES
0	5472 MB	1,11,58,912	8
		approx · all schemas	logical

Host machine

CPU / RAM where Postgres + Node ran

METRIC	VALUE
Hostname	LAPTOP-0VLFL1GO
OS	Windows_NT 10.0.19045 (x64)
CPU	Intel(R) Core(TM) i5-9300H CPU @ 2.40GHz
Logical cores	8
CPU clock (avg)	2400 MHz
RAM	7.9 GB total · 1.8 GB free
Node.js	v24.15.0

PostgreSQL settings

Local GlowTap instance

METRIC	VALUE
Database	GlowTap
Version	18.4
shared_buffers	128 MB
work_mem	64 MB
effective_cache_size	4096 MB
max_parallel_workers	8
max_parallel_workers_per_gather	4
max_connections	100

Local Postgres — CPU/RAM below are from this machine (client and DB host).

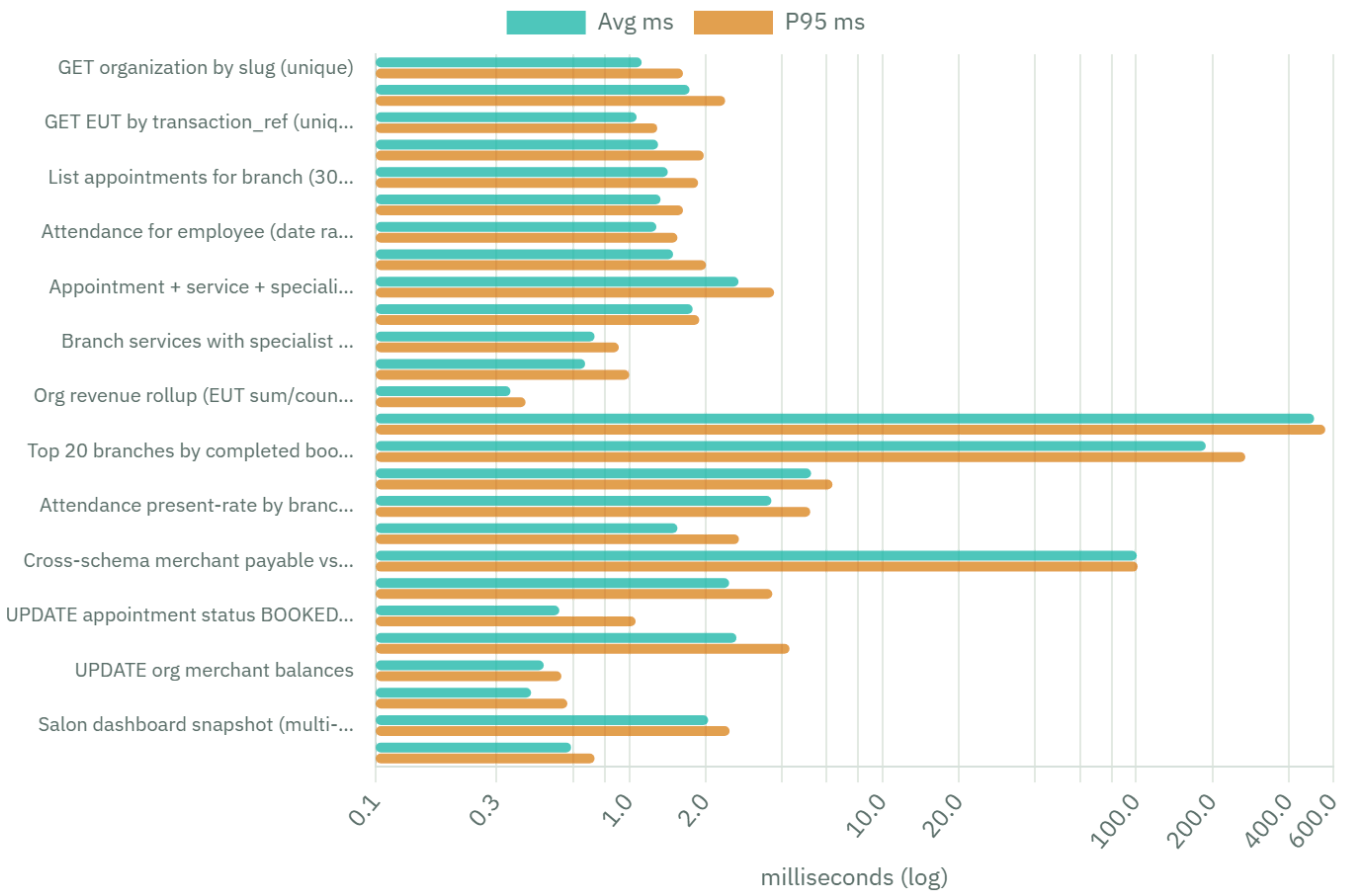
Dataset under test

PERF seed + tracked live rows

METRIC	VALUE
Database size	5472 MB
Approx. live rows	1,11,58,912
PERF seed row sum	86,02,500
Organizations	2,500
Branches	12,500
Appointments	25,00,000
Status events	25,00,000
End-user transactions	8,62,500
Attendance rows	25,00,000
Employees	1,00,000
Branch services	1,25,000

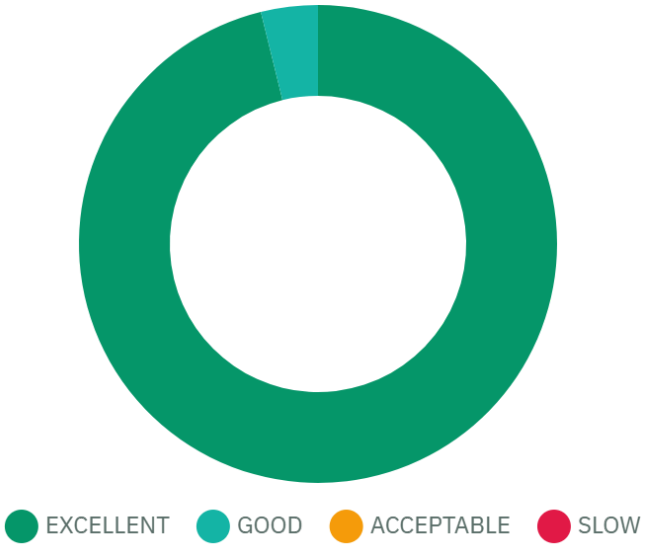
Latency spectrum

Avg vs p95 across every benchmark (log scale)



Grade mix

Share of results under local SLO bands



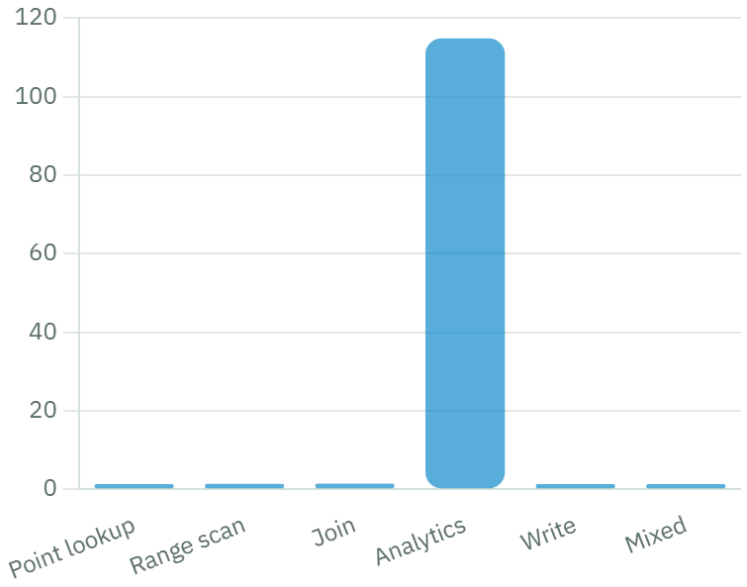


Suite health

Weighted from grades (Excellent=4 … Slow=0). Target: keep analytics under 1s at this scale.

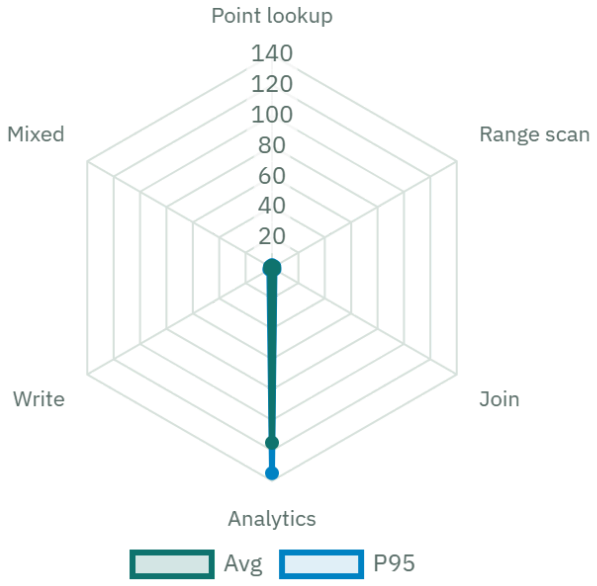
By category

Average latency



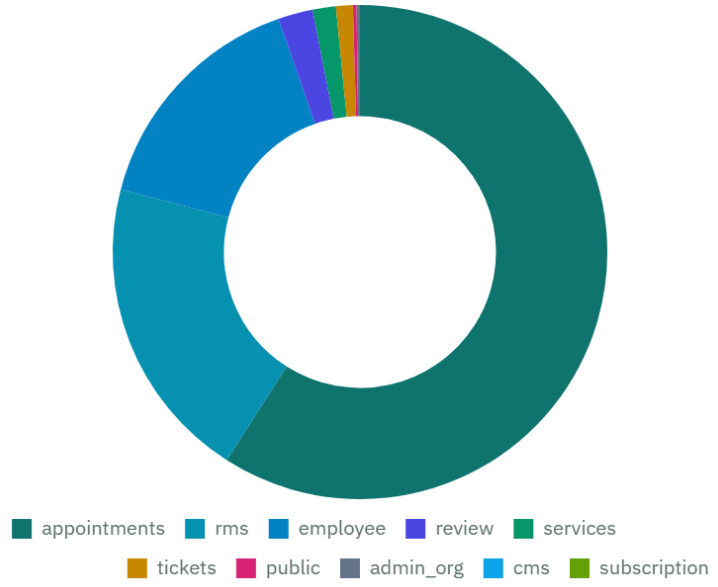
Category radar

Avg vs p95 profile



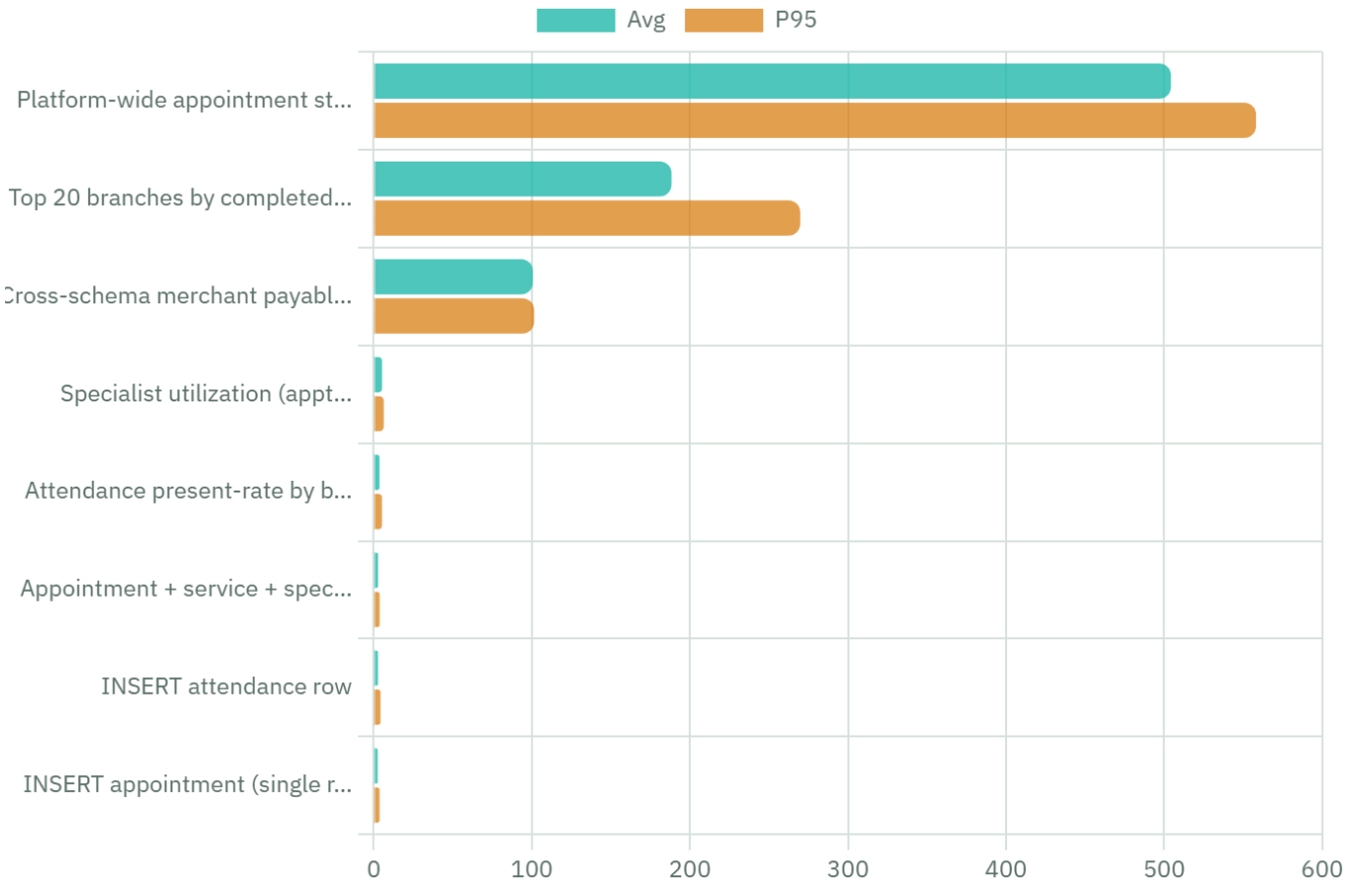
Storage by schema

Disk footprint (MB)



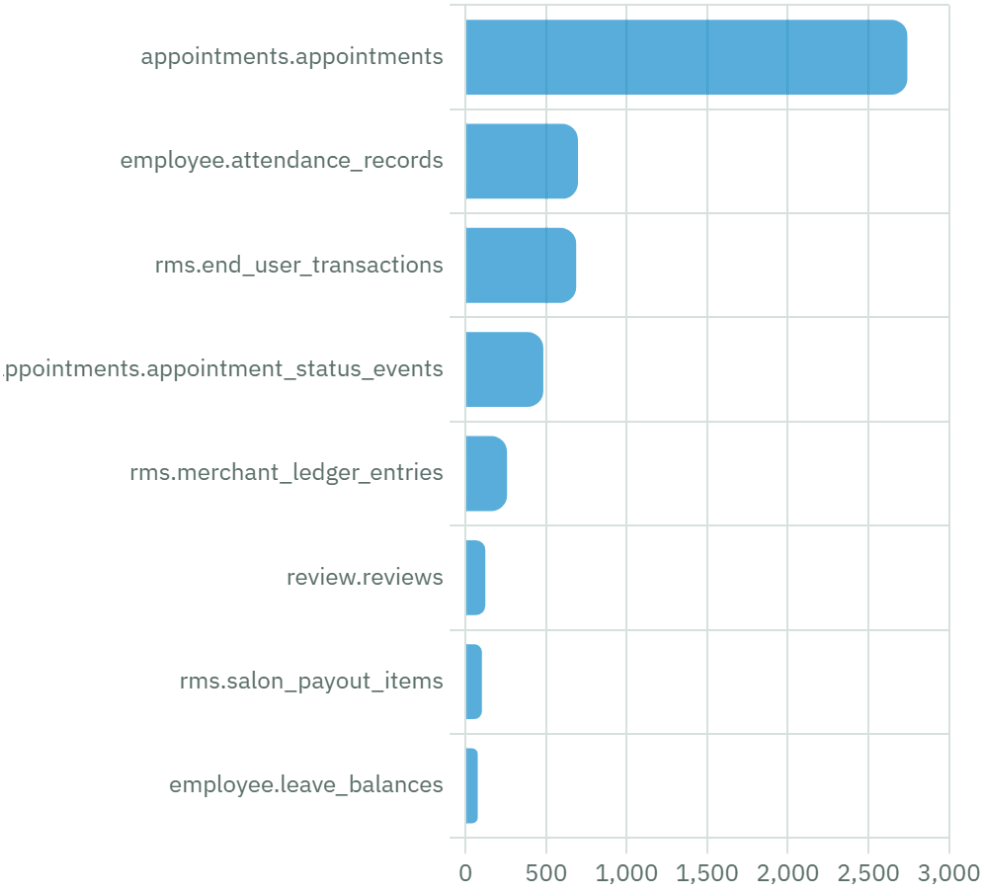
Slowest queries

Highest avg ms after warm-up



Largest tables

Live rows × size



Results matrix

All timings in milliseconds · sticky header for scanning

TEST	CATEGORY	ROWS	AVG	P50	P95	MAX		G
GET organization by slug (unique)	Point lookup	1	1.12	1.03	1.63	1.63	—	EXCELL
GET appointment by booking_number (unique)	Point lookup	1	1.73	1.44	2.39	2.39	—	EXCELL
GET EUT by transaction_ref (unique)	Point lookup	1	1.07	1.06	1.29	1.29	—	EXCELL
GET employee by org + employee_number	Point lookup	0	1.3	1.27	1.97	1.97	—	EXCELL
List appointments for branch (30-day window)	Range scan	0	1.42	1.22	1.87	1.87	—	EXCELL
Filter appointments by status + org	Range scan	0	1.33	1.19	1.63	1.63	—	EXCELL

TEST	CATEGORY	ROWS	AVG	P50	P95	MAX	G

Hottest relations

Top tables by on-disk size

RELATION	LIVE ROWS	SIZE
appointments.appointments	24,84,835	2742 MB
employee.attendance_records	24,87,809	698 MB
rms.end_user_transactions	8,64,394	687 MB
appointments.appointment_status_events	25,00,045	483 MB
rms.merchant_ledger_entries	8,62,500	258 MB
review.reviews	2,50,003	123 MB
rms.salon_payout_items	4,37,500	102 MB
employee.leave_balances	3,00,006	76 MB
services.branch_services	1,25,012	46 MB
employee.employees	1,00,002	43 MB

Methodology

Client wall time via `perf_hooks` around `pg` queries. One warm-up discarded; stats from 5 rounds (min / avg / p50 / p95 / p99 / max). Covering + partial indexes (`076` / `077`), range predicates, and LATERAL aggregates for platform analytics. Local SLO bands are tighter than Neon (no WAN RTT).

▶ **Full markdown appendix**

GlowTap — Database Performance · generated by scripts/run-perf-benchmark.js